



Analysis on IT Skills & Trends

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- In the competitive global IT sector, it is essential to keep up with the ever changing technologies. This study uses Data analytics to highlight Current and Future skills required towards enhancement of IT Career based on the Survey conducted in the year 2019.
- Data sources from
 - Stack Overflow Survey
 - IBM Site
 - Github Job Postings
- The finding shows what are all the current and future trending programming languages, databases and Web frame technologies around the world.
- It also provides demographical view of Technologies which helps in terms of Hiring and Training process.

INTRODUCTION



AIMS GRID

- **Purpose:**
 1. What are all the Current and Future Top Technologies preferred mostly?
 2. and Demographical Identification of Technology usages
- **Stakeholders:**

HR, IT Heads, Job Hoppers
- **Criteria:**

Based on Most preferences by the established developers' experiences and interest

METHODOLOGY



Using Python:

- Data Collection using Web scrapping
- Data Exploration
 - Data Cleaning
 - Data Wrangling
- Visualization of Data Distribution, Relationship, and Comparision
- Finding Correlations

Using IBM Cognos:

- Dashboard Creation

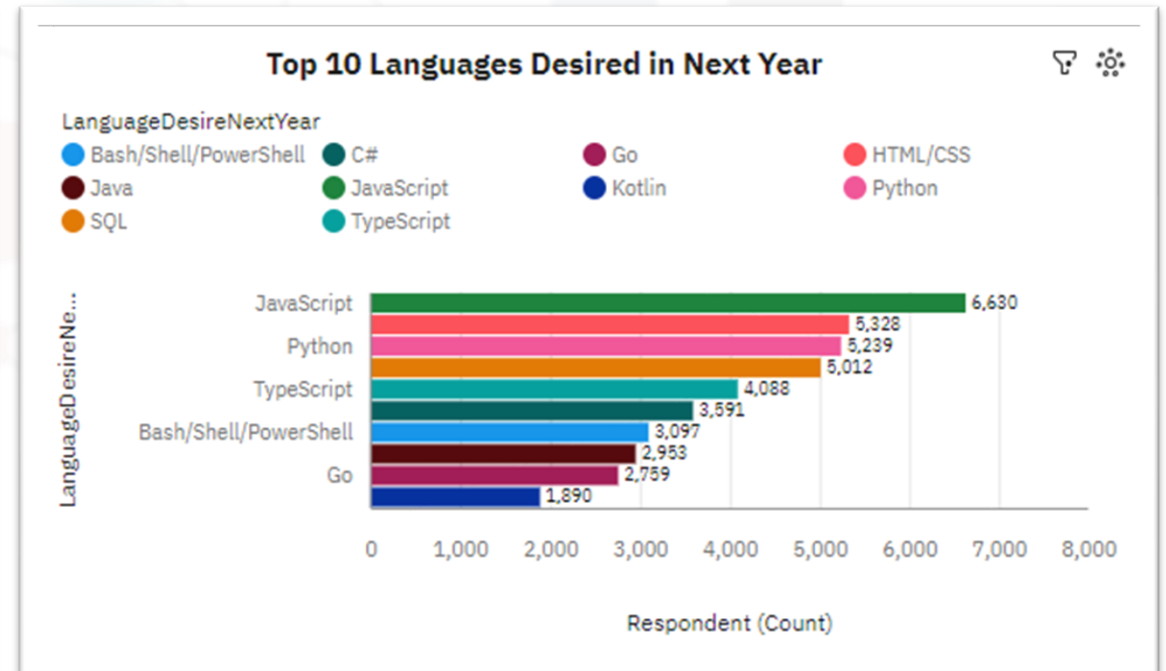
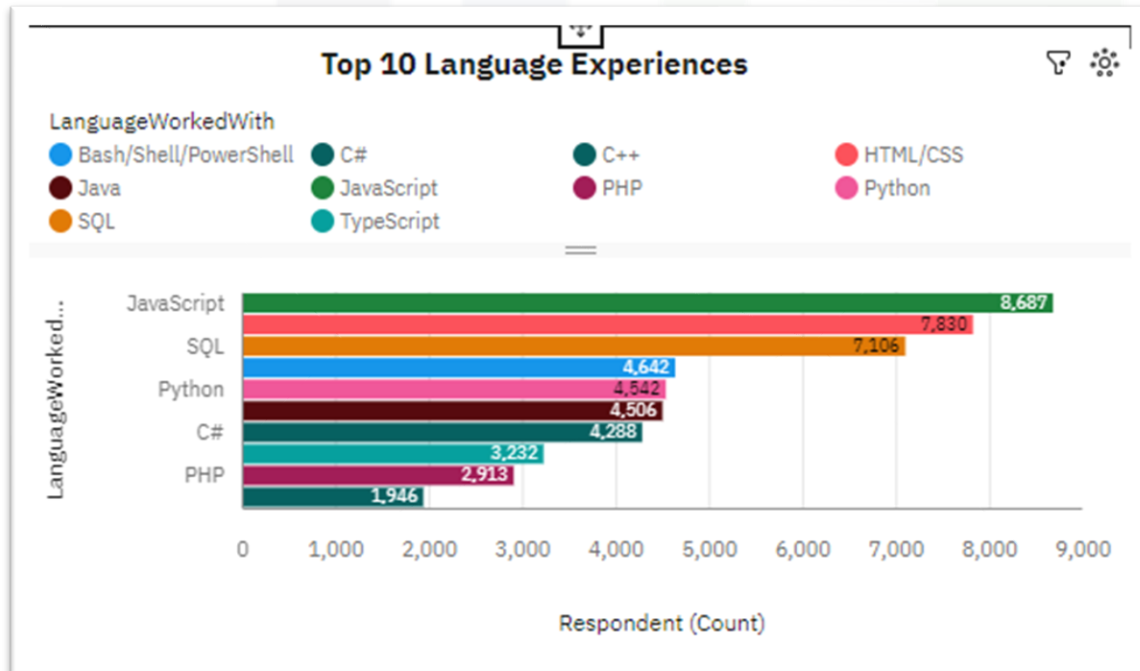
RESULTS



PROGRAMMING LANGUAGE TRENDS

Current Year

Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

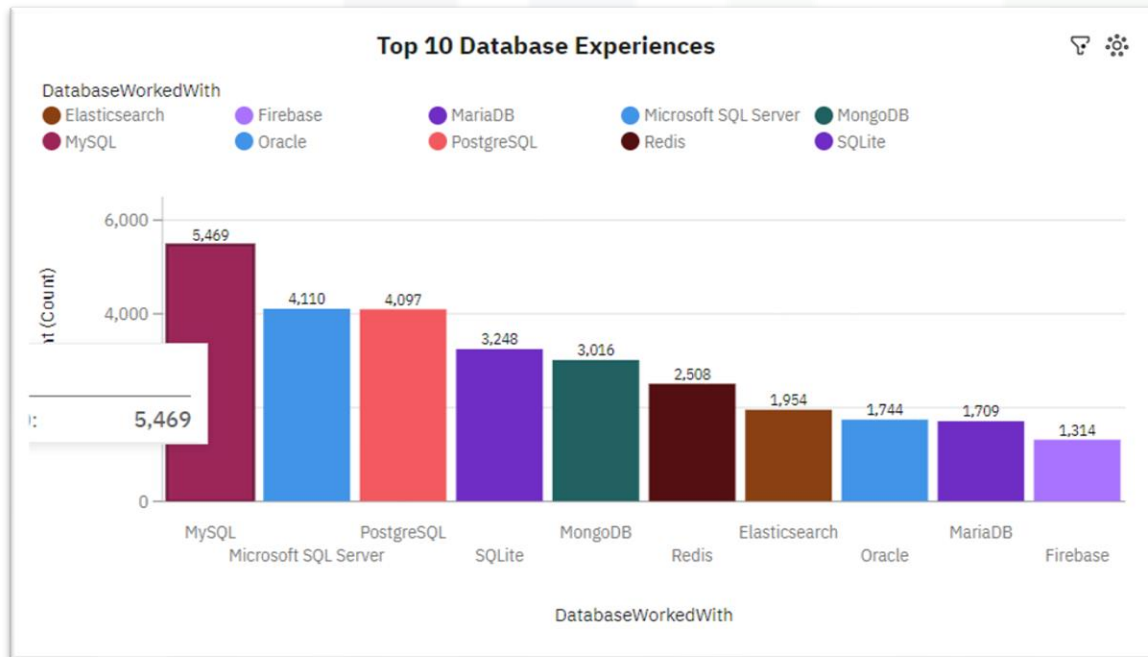
- Web development Languages JavaScript and HTML/CSS are dominant in current and future
- Python is emerging due to developing environments of Data Science and AI fields
- Go has got future attention as highly preferred language for AI development

Implications

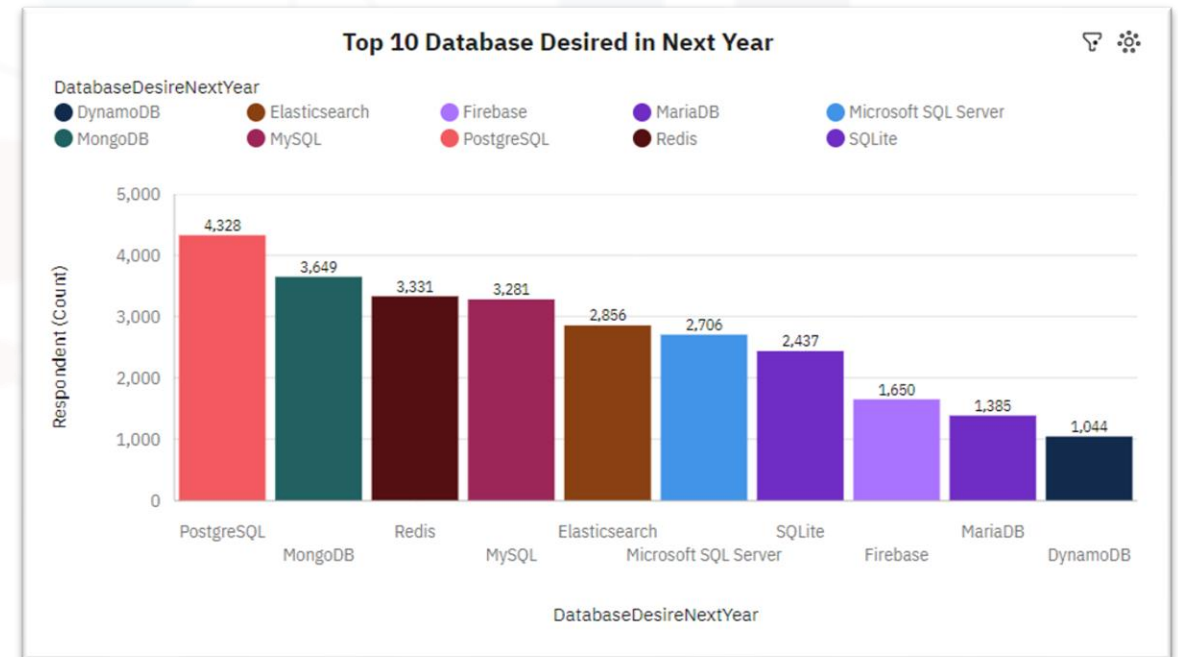
- It shows that JavaScript and HTML/CSS has good potential developers in future
- Importance towards Python for future consideration of Hiring and Training
- Languages helpful on AI development given more important

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- There is significant drop in MySQL and Microsoft SQL Server
- Increased interest in Postgre SQL, Mongo DB and Redis
- Oracle dropped from Top 10 list in future expectations

Implications

- Investment shift addition towards Postgre SQL, Mongo DB and Redis are suggested
- SQL tools are still preferred by many respondents
- Need to reconsider Oracle Training and Hiring for future

DASHBOARD

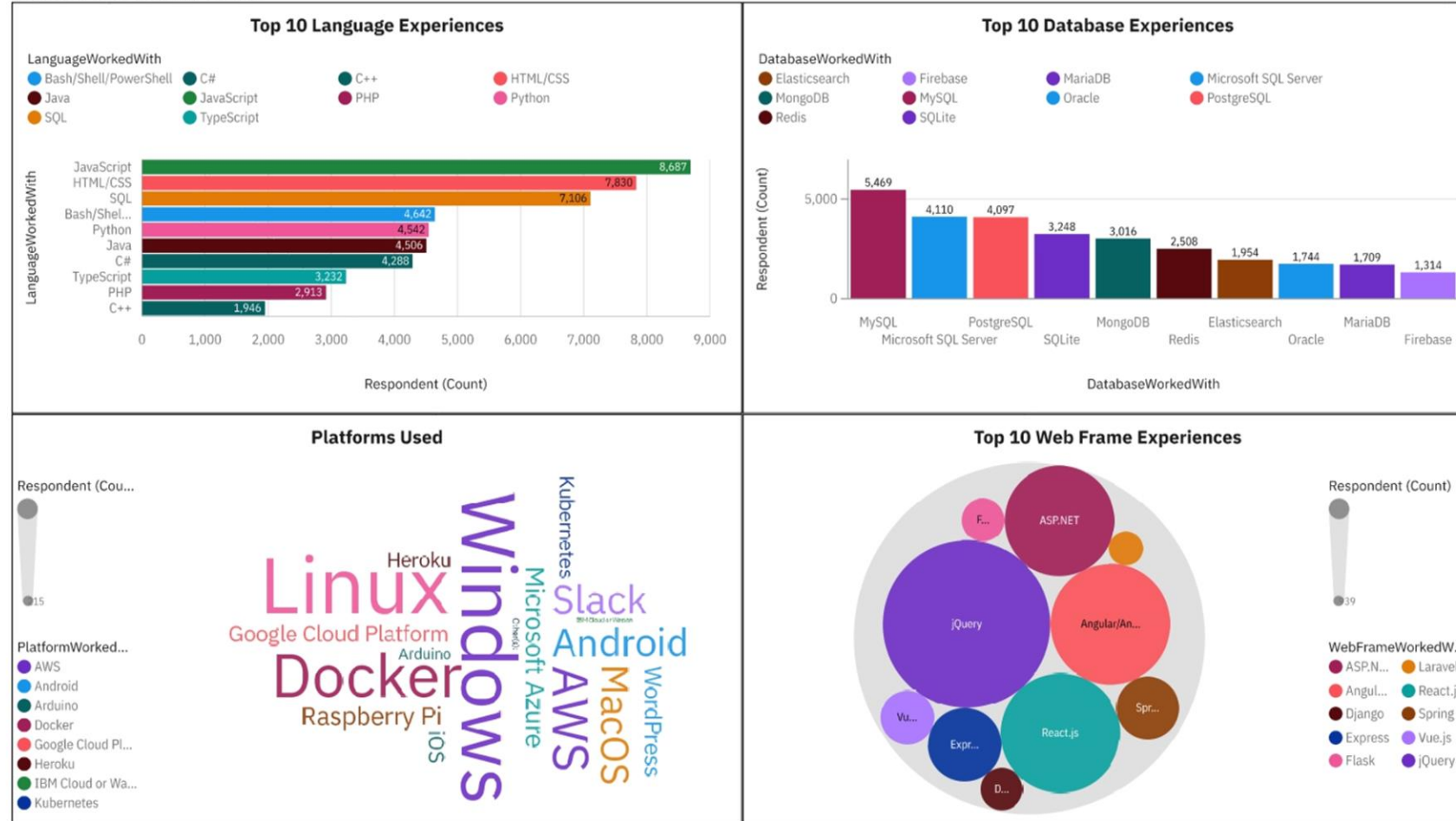


[Github Link](#)

To open kindly press Ctrl + click

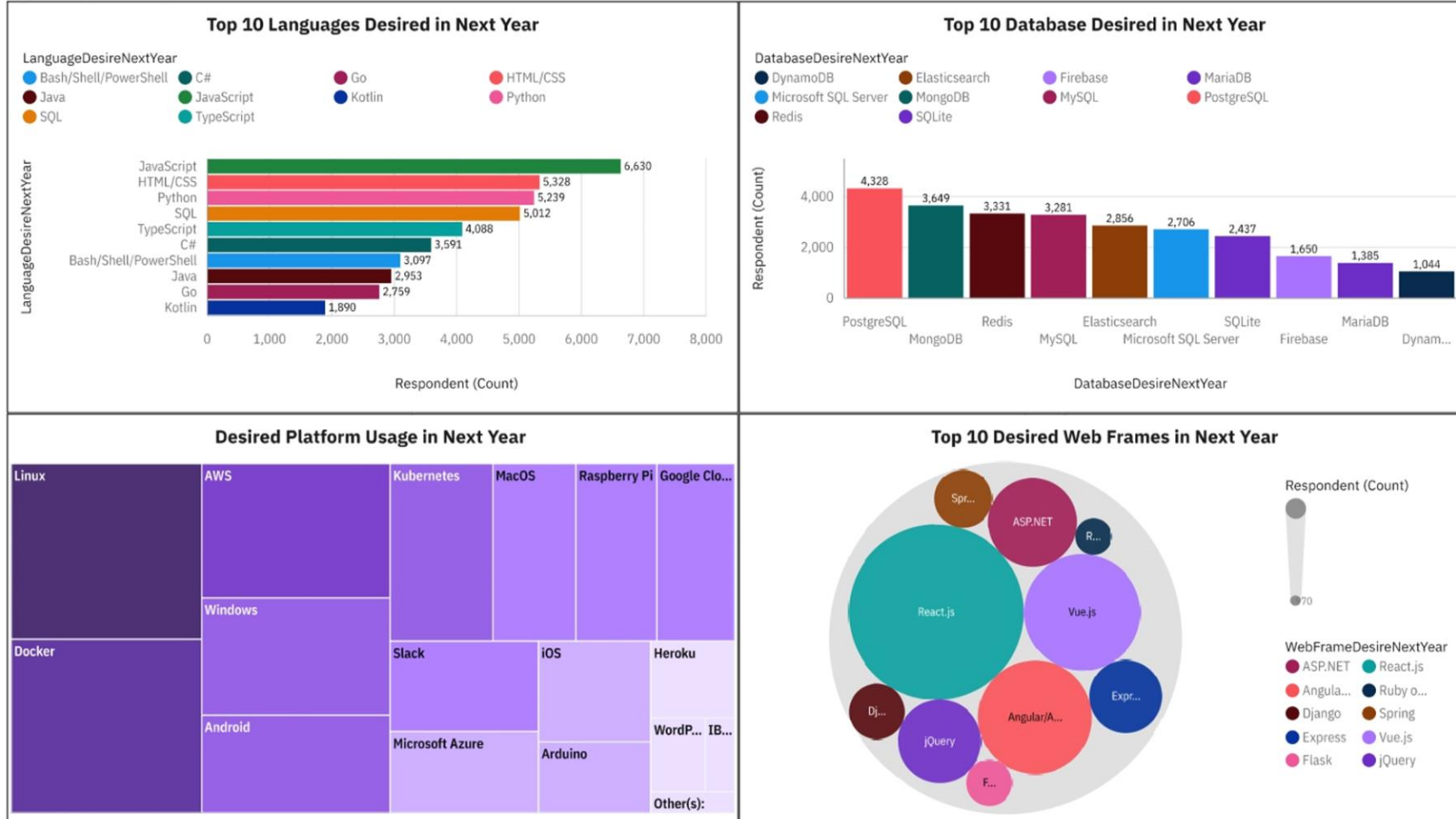
DASHBOARD TAB 1

Current Technology Usage

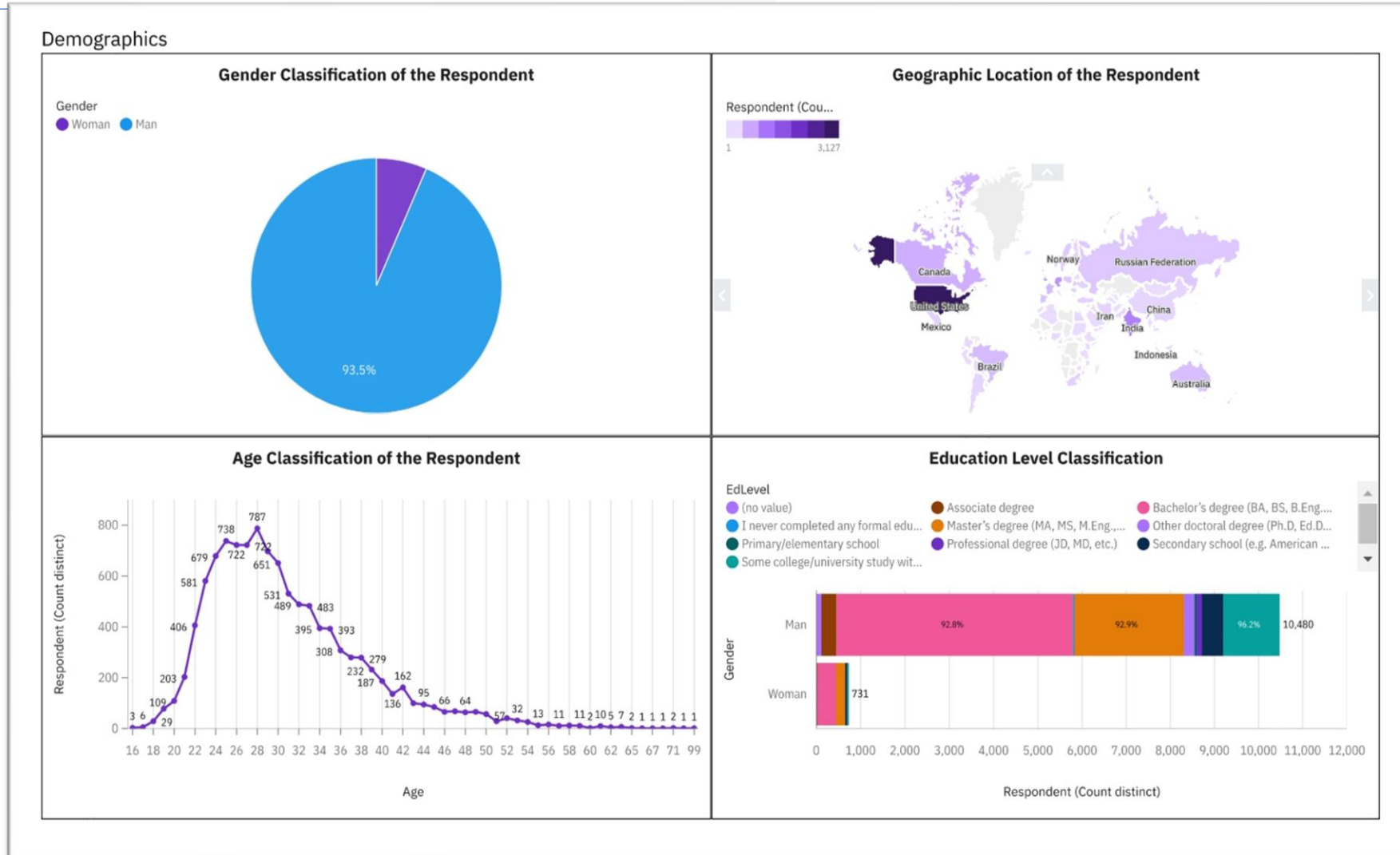


DASHBOARD TAB 2

Future Technology Trend



DASHBOARD TAB 3



DISCUSSION



- Upskilling in Technology sector
- How do we close the wide Gender gap?
- More Tech education, access and development in less development regions in South East Asia, South America, and some part of Europe
- Relevancy of lower interested Technology like Oracle, C++ etc

OVERALL FINDINGS & IMPLICATIONS

Findings

- Most people in IT field have a Bachelor degree
- Web development languages are the most popular in the IT sector
- Tech sector are filled with the age group of people under 40 years
- Most respondents wants to learn Postgre SQL and react JS next year

Implications

- It is important for data professionals to develop proficiency in NoSQL in addition to SQL databases
- Web development is still a very lucrative skill
- Less developed countries needs more access to training and education in IT sector.

CONCLUSION



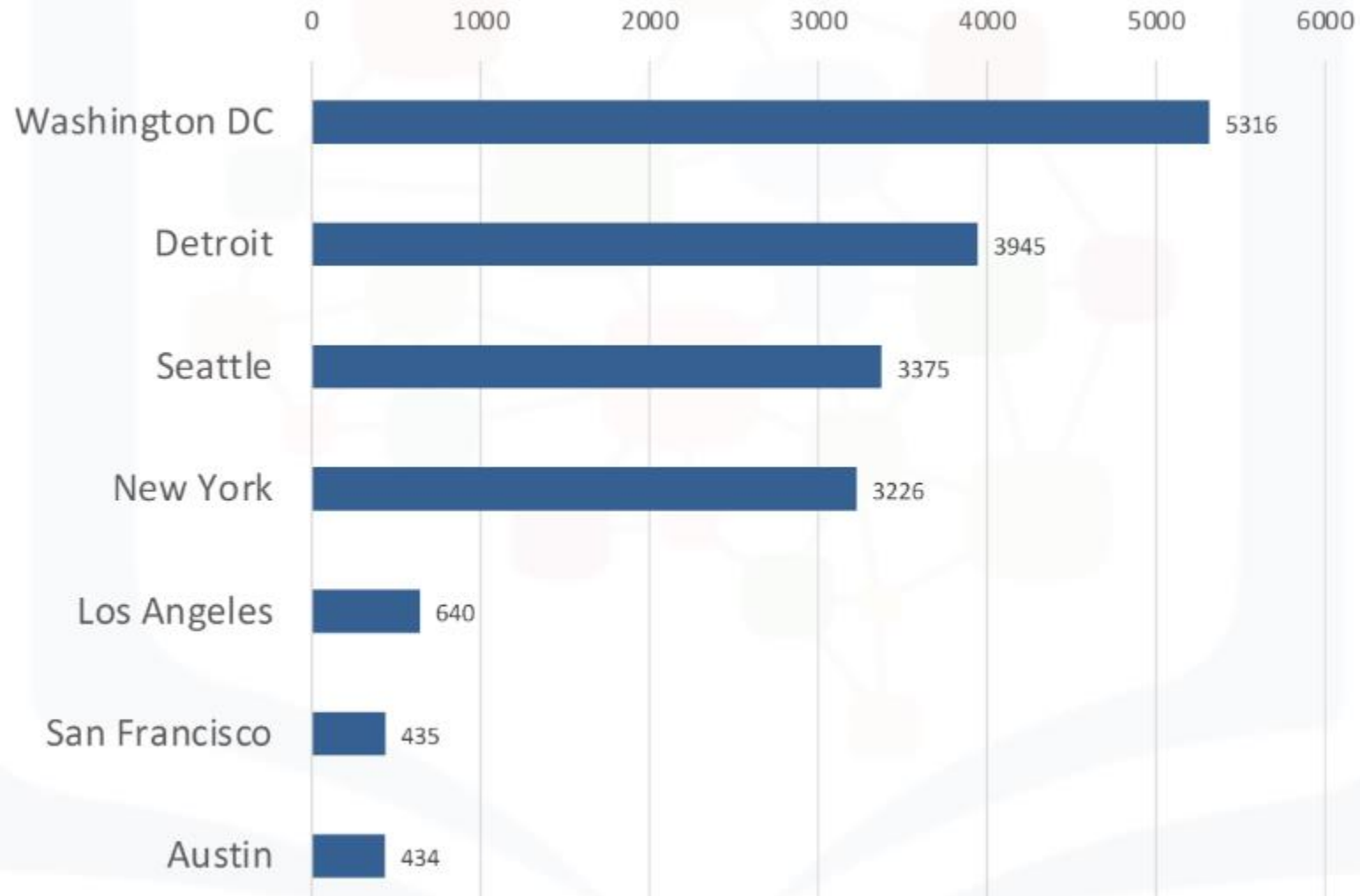
- The findings underscore the dynamic nature of the programming landscape and the critical role of technology in driving innovation across industries
- As developers navigate this ever changing terrain with a keen understanding of diverse programming languages. And Database systems become essential to meet the demand of modern applications

APPENDIX



1. Job Postings
2. Popular Languages

1. JOB POSTINGS



2. POPULAR LANGUAGES

